

Project 2 — Estimating the Truth

Building high-quality estimates from samples.

Project 2: Estimation from a simple random sample

Total points: 35

Due: Sunday, March 29 (Week 9; see [Assessments](#) and Canvas)

In this project, you will use a **95% confidence interval** to estimate a **population proportion** or **population mean** using data collected from a **simple random sample**. Complete the steps below, then submit a **one-page summary** and an **R notebook** that runs top to bottom.

Sample write-up: [Download sample one-page write-up — CA high schools \(PDF\)](#). HTML version: [Project 2 — Example: mean age of California high schools](#). Do not copy my population or variable — write up your own study following the grading rubric below.

Sample R notebooks (adapt to your data; choose **mean** or **proportion**):

Goal	Notebook	Open in Colab
Estimate a population mean with FPC	project-2-estimation-mean-fpc.ipynb	Open in Colab
Estimate a population proportion with FPC	project-2-proportion-fpc.ipynb	Open in Colab

After opening in Colab, use **Runtime** → **Change runtime type** and select **R** if needed.

Important dates

Date	Milestone
Lab week	R code and CI types in Module 2 demo lab · exercise lab
Thursday, March 26	Peer review of sampling method and interval choice in lab
Sunday, March 29	One-page summary and R notebook due in Canvas

What to turn in

1. **One-page summary (PDF or equivalent)** — organized using the section headings in the rubric below.
2. **R notebook** (.ipynb, .qmd, or .Rmd) — all code to load your data, make figures, and compute your 95% confidence interval. The notebook must run from top to bottom.

Steps

1. Choose a finite population you can list

Think of a **population you could list**, randomly sample from the list, and visit (or inspect) each sampled item to collect **one variable** — either a **number** or a **binary outcome**.

It is hard to randomly sample **people** without university research approval and without a complete contact list for the population. For these reasons, I recommend a population of **non-human objects**. For example:

- Number all parking spaces in a lot (e.g. from a Google Maps image), randomly select 10 spaces, and record one variable at each space (occupied or not? truck or not? white or not? model year?).
- Sample from a numbered list of items on a website, as in the [sample project](#).

Your turn.

My population is _____, which has $N =$ _____ objects.

2. Choose the parameter and variable

Once you have a population, decide what **mean** or **proportion** you want to estimate. Identify the **variable** you will collect for each sampled object.

Your turn.

My variable is _____, which is ☐ binary ☐ categorical ☐ numerical (circle one), so I'll be estimating a ☐ proportion ☐ mean (circle one).

3. Choose a sample size

Larger samples give **more precision**, but collecting data takes time. In professional work, a **sample size calculation** balances time and precision. For this project, collect data from **10 objects** or the **largest number** you can feasibly visit in **1–2 hours** — whichever you choose as your sample size.

Your turn.

I think the largest number of objects I could feasibly visit and collect information for is $n =$ _____.

4. Draw a simple random sample

Use a **random number generator** to select n distinct items from 1 to N . We have used this one in class:

[Rossman Chance random number generator](#)

Your turn.

The n items I need to visit are: _____

5. Collect your data

Visit each sampled item and record your variable. Keep your dataset in R (a vector or a small data frame).

Your turn.

The dataset from my sample is: _____

6. Compute your point estimate

Compute the **sample mean** (numerical variable) or **sample proportion** (binary variable). This is your single best estimate of the population parameter.

Your turn.

My single best estimate of the population is _____

7. Choose and compute a 95% confidence interval in R

Decide which of the **five CI situations** from Read 1 and the demo lab applies to your data:

3. Results

3a. Descriptive statistics (6 pts)

- Include a **well-labelled graph** of your dataset.
- Include the **mean and SD** if the variable is numerical, or the **count and proportion** if the variable is binary categorical.

3b. Inferential statistics (6 pts)

- Include the values of L and U from your 95% confidence interval.
- Include a **one-sentence interpretation** of the interval in the context of your parameter and population.

4. Conclusion (6 pts)

- Briefly describe your **reaction** to the final estimate — is it about what you expected?
- Briefly state whether you believe your estimate is **biased**.
- Briefly state how you could make your estimate **more precise**.